myWMS Stack: WildFly 22 + PostgreSQL 16.9 + ReportServer — Complete Guide

Preamble (disclaimer)

This is an internal technical guide. Examples, commands, and configurations are provided **AS IS**, without any warranty of accuracy or fitness. Use at your own risk.

- **Before running:** verify values in .env, paths, permissions, and ports; review docker-compose.yml.
- **Versions/environments:** differences in Docker/Compose/OS/JDK/PostgreSQL/WildFly/Tomcat may require adjustments.
- **Security:** the sample pg_hba.conf is permissive for first boot; restrict networks and rotate passwords for production.
- **Backups:** back up data before major changes/upgrades.
- **Startup order:** postgres → (optional rs_init) → mywms & reportserver. Shutdown in reverse order.

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1) Goal & architecture

- **PostgreSQL 16.9** — databases mywms & reportserver, users mywms & reportserver.
- **myWMS** on **WildFly 22** (Java 11) — local WF22 tree + standalone.xml; deployments/ contains mywms.ear and postgresql-42.2.19.jar.
- **ReportServer CE** (Tomcat 9 + JDK 11) — unpacked webapp/reportserver tree; DB schema is imported *before* first start.

myWMS DS/JNDI: DS_NAME=PostgreSQLPool, JNDI_NAME=java:/losDS. JDBC: jdbc:postgresql://pg:5432/mywms.

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2) Requirements & versions

- OS: Ubuntu 20.04 / 22.04 / 24.04
- Docker & Compose
- Java: Temurin 11
- WildFly: 22.0.1.Final (local unpacked directory)
- PostgreSQL: 16.9
- ReportServer CE: RS 4.7.7

Minimum HW: 2 vCPU, 8 GB RAM. For ReportServer: 4 vCPU, 16 GB RAM recommended.

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3) Directory layout

```
mywms-stack/
├─ .env.example
├─ .env
├─ docker-compose.yml
├─ postgres/
│  ├─ conf/
│  │  ├─ pg_hba.conf
```

```
│   │   └─ postgresql.conf
│   ├─ init/
│   │   ├─ 01-create-db-users.sql
│   │   ├─ 02-load-mywms.sh
│   │   └─ rs_post_init.sql
│   └─ backup/
│       ├─ backup-all.sh
│       ├─ backup-mywms.sh
│       ├─ backup-reportserver.sh
│       └─ restore-mywms.sh
├─ mywms/
│   ├─ Dockerfile
│   ├─ standalone.xml
│   └─ deployments/
│       ├─ mywms.ear
│       └─ postgresql-42.2.19.jar
└─ reportserver/
    ├─ Dockerfile
    ├─ webapp/
    │   └─ reportserver/
    │       ├─ WEB-INF/
    │       │   └─ classes/
    │       │       ├─ jasperreports.properties
    │       │       ├─ jasperreports_extension.properties
    │       │       └─ fonts/
    │       │           ├─ dejavu.xml
    │       │           └─ ttf/ (DejaVuSans*.ttf)
    │       └─ ddl/
    │           └─ reportserver-RS4.7.7-6117-schema-PostgreSQL_CREATE.sql
    └─ config/
        ├─ persistence.properties
        └─ reportserver.properties
```

4) Top-level files & content

4.1 .env.example

```
# PostgreSQL superuser
POSTGRES_USER=postgres
POSTGRES_PASSWORD=postgres

# Application databases & users
MYWMS_DB=mywms
MYWMS_USER=mywms
MYWMS_PASS=mywms

RS_DB=reportserver
RS_USER=reportserver
RS_PASS=reportserver

# Ports
PG_PORT=5432
MYWMS_HTTP=8080
MYWMS_MGMT=9990
RS_HTTP=8085
PGADMIN_HTTP=5050

# Host paths for persistence/backups
PG_HOST_DATA=/srv/pgadata
PG_HOST_BACKUP=/srv/pgbackup
```

4.2 docker-compose.yml

```
services:
  postgres:
    image: postgres:16.9
    container_name: pg
    environment:
      POSTGRES_USER: ${POSTGRES_USER}
      POSTGRES_PASSWORD: ${POSTGRES_PASSWORD}
    ports:
      - "${PG_PORT}:5432"
```

```
    volumes:
      - "${PG_HOST_DATA}:/var/lib/postgresql/data"
      - "${PG_HOST_BACKUP}:/backups"
      - "./postgres/conf/pg_hba.conf:/etc/postgresql/pg_hba.conf:ro"
      - "./postgres/conf/postgresql.conf:/etc/postgresql/postgresql.conf:ro"
      - "./postgres/init:/docker-entrypoint-initdb.d:ro"
    command: ["postgres", "-c", "config_file=/etc/postgresql/postgresql.conf"]
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -U ${POSTGRES_USER} -d postgres"]
      interval: 5s
      timeout: 5s
      retries: 30

  rs_init:
    image: postgres:16.9
    depends_on:
      postgres:
        condition: service_healthy
    entrypoint: ["/bin/bash", "-lc"]
    working_dir: /ddl
    volumes:
      - "./reportserver/webapp/reportserver/ddl:/ddl:ro"
    command: >
      "
      psql -h pg -U ${POSTGRES_USER} -d ${RS_DB} -f /ddl/reportserver-RS4.7.7-6117-schema-
PostgreSQL_CREATE.sql &&
      psql -h pg -U ${POSTGRES_USER} -d ${RS_DB} -f /ddl/rs_grants.sql
      "
    environment:
      PGPASSWORD: ${POSTGRES_PASSWORD}

  mywms:
    build:
      context: ./mywms
    container_name: mywms
    depends_on:
      postgres:
        condition: service_healthy
    ports:
```

```
      - "${MYWMS_HTTP}:8080"
      - "${MYWMS_MGMT}:9990"
    volumes:
      - "./mywms/deployments:/opt/wildfly/standalone/deployments:rw"

  reportserver:
    build:
      context: ./reportserver
    container_name: reportserver
    depends_on:
      postgres:
        condition: service_healthy
    ports:
      - "${RS_HTTP}:8080"
    environment:
      - TZ=Europe/Belgrade
    volumes:
      - "./reportserver/config:/config:rw"
      - "rs_lucene:/lucene_index"

volumes:
  rs_lucene: {}
```

---

5) PostgreSQL — base initialization

5.1 postgres/conf/pg_hba.conf

```
local   all         all                         trust
host    all         all         0.0.0.0/0       md5
host    all         all         ::/0            md5
```

5.2 postgres/conf/postgresql.conf

```
listen_addresses = '*'
timezone = 'Europe/Belgrade'
```

5.3 postgres/init/01-create-db-users.sql

```
DO $$
BEGIN
  IF NOT EXISTS (SELECT FROM pg_roles WHERE rolname='mywms') THEN
    CREATE ROLE mywms LOGIN PASSWORD 'mywms';
  END IF;
  IF NOT EXISTS (SELECT FROM pg_roles WHERE rolname='reportserver') THEN
    CREATE ROLE reportserver LOGIN PASSWORD 'reportserver';
  END IF;
END $$;

DO $$
BEGIN
  IF NOT EXISTS (SELECT FROM pg_database WHERE datname='mywms') THEN
    EXECUTE 'CREATE DATABASE mywms OWNER mywms';
  END IF;
  IF NOT EXISTS (SELECT FROM pg_database WHERE datname='reportserver') THEN
    EXECUTE 'CREATE DATABASE reportserver OWNER reportserver';
  END IF;
END $$;
```

5.4 postgres/init/02-load-mywms.sh

```
#!/usr/bin/env bash
set -euo pipefail
DB="${MYWMS_DB:-mywms}"
DUMP="/docker-entrypoint-initdb.d/mywms"
if [[ -f "$DUMP" ]]; then
  echo "[INIT] Detecting dump format: $DUMP"
  if file "$DUMP" | grep -qi "PostgreSQL custom-format dump"; then
    echo "[INIT] Custom dump detected → pg_restore --no-owner --no-privileges --role=mywms"
    pg_restore -U postgres -d "$DB" --no-owner --no-privileges --role=mywms "$DUMP"
  else
    echo "[INIT] Plain SQL → psql -f"
    psql -U postgres -d "$DB" -f "$DUMP"
  fi
else
  echo "[INIT] No mywms dump provided (optional)"
```

```
fi
```

5.5 postgres/init/rs_post_init.sql

```
GRANT CONNECT ON DATABASE reportserver TO reportserver;
GRANT USAGE ON SCHEMA public TO reportserver;
GRANT SELECT, INSERT, UPDATE, DELETE ON ALL TABLES IN SCHEMA public TO reportserver;
GRANT USAGE, SELECT, UPDATE ON ALL SEQUENCES IN SCHEMA public TO reportserver;
ALTER DEFAULT PRIVILEGES IN SCHEMA public GRANT SELECT, INSERT, UPDATE, DELETE ON TABLES TO reportserver;
ALTER DEFAULT PRIVILEGES IN SCHEMA public GRANT USAGE, SELECT, UPDATE ON SEQUENCES TO reportserver;
```

---

6) myWMS — WildFly 22

6.1 mywms/Dockerfile

```
FROM eclipse-temurin:11-jre
ADD wildfly-22.0.1.Final /opt/wildfly
ENV JBOSS_HOME=/opt/wildfly
WORKDIR ${JBOSS_HOME}
COPY standalone.xml ${JBOSS_HOME}/standalone/configuration/standalone.xml
EXPOSE 8080 9990
ENTRYPOINT ["bash","-lc","${JBOSS_HOME}/bin/standalone.sh -b 0.0.0.0 -bmanagement 0.0.0.0"]
```

6.2 standalone.xml (key points)

- JDBC URL: jdbc:postgresql://pg:5432/mywms
- DS/JNDI: PostgreSQLPool / java:/losDS
- Driver: postgresql-42.2.19.jar placed in standalone/deployments/

6.3 deployments/

mywms.ear and postgresql-42.2.19.jar (add .dodeploy marker if needed).

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7) ReportServer — Tomcat

7.1 Layout

Unpacked webapp/reportserver under reportserver/webapp/.

7.2 reportserver/Dockerfile

```
FROM tomcat:9-jdk11-temurin
ENV CATALINA_OPTS="-Xms1g -Xmx2g -Djava.net.preferIPv4Stack=true -Drs.configdir=/config -
Drs.lucene.dir=/lucene_index"
ENV LANG=en_US.UTF-8 TZ=Europe/Belgrade
COPY webapp/reportserver /usr/local/tomcat/webapps/reportserver
RUN mkdir -p /config /lucene_index
EXPOSE 8080
CMD ["catalina.sh","run"]
```

7.3 config/persistence.properties

```
hibernate.dialect=net.datenwerke.rs.utils.hibernate.PostgreSQLDialect
hibernate.connection.driver_class=org.postgresql.Driver
hibernate.connection.url=jdbc:postgresql://pg:5432/reportserver
hibernate.connection.username=reportserver
hibernate.connection.password=reportserver
hibernate.default_schema=public
hibernate.hbm2ddl.auto=validate
```

7.4 config/reportserver.properties

```
rs.configdir=/config
```

7.5 Schema import (rs_init)

```
docker compose up -d postgres
docker compose run --rm rs_init
docker compose up -d mywms reportserver
```

7.6 Cyrillic (Unicode) in PDF/HTML (Jasper)

Ensure fonts and mapping files are present inside the container (copied via image build):

```
WEB-INF/classes/fonts/ttf/DejaVuSans.ttf
WEB-INF/classes/fonts/ttf/DejaVuSans-Bold.ttf
WEB-INF/classes/fonts/dejavu.xml

# dejavu.xml



  fonts/ttf/DejaVuSans.ttf
  fonts/ttf/DejaVuSans-Bold.ttf
  fonts/ttf/DejaVuSans.ttf
  fonts/ttf/DejaVuSans-Bold.ttf
  Identity-H
  true




# jasperreports_extension.properties
net.sf.jasperreports.extension.registry.factory.fonts=net.sf.jasperreports.engine.fonts.SimpleFontExtensio
nsRegistryFactory
net.sf.jasperreports.extension.simple.font.families.dejavu=fonts/dejavu.xml
```

7.7 Clear Tomcat work/temp (cache)

```
docker compose stop reportserver
docker compose rm -f reportserver
docker compose up -d reportserver
# or
docker exec -it reportserver bash -lc 'rm -rf /usr/local/tomcat/work/* /usr/local/tomcat/temp/*'
docker compose restart reportserver
```

---

8) PostgreSQL tuning

```
shared_buffers = 2GB
effective_cache_size = 6GB
maintenance_work_mem = 1GB
work_mem = 64MB
max_connections = 200
```

```
autovacuum_naptime = 20s
autovacuum_vacuum_scale_factor = 0.1
autovacuum_analyze_scale_factor = 0.05
```

---

9) Useful commands

```
# Bring up
docker compose up -d postgres
docker compose run --rm rs_init
docker compose up -d mywms reportserver

# Stop (reverse order)
docker compose stop reportserver mywms
docker compose stop postgres

# Logs
docker compose logs -f postgres
docker compose logs -f mywms
docker compose logs -f reportserver

# PostgreSQL checks
docker exec -it pg psql -U postgres -d postgres -c "\l"
docker exec -it pg psql -U postgres -d postgres -c "\du"
docker exec -it pg psql -U postgres -d reportserver -c "\dt"
docker exec -it pg psql -U postgres -d mywms -c "SELECT now(), current_user;"
```

---

10) Updating mywms.ear

```
cp NEW_mywms.ear mywms/deployments/mywms.ear
touch mywms/deployments/mywms.ear.dodeploy
docker compose restart mywms
```

---

11) Troubleshooting

- **postgresql.conf error:** verify minimal syntax/values.
- **DO $$ blocks:** ensure plain $$ (no backslashes).
- **RS missing tables:** run rs_init (schema), then rs_post_init.sql grants.
- **RS permission denied:** re-apply grants (see 5.5).
- **Jasper Cyrillic:** fonts + config + clear Tomcat cache.

---

12) Persistence, backups, security

```
# backup-all.sh
#!/usr/bin/env bash
set -euo pipefail
ts=$(date +%F_%H%M%S)
pg_dump -U postgres -d mywms -Fc -f /backups/mywms_$ts.dump
pg_dump -U postgres -d reportserver -Fc -f /backups/reportserver_$ts.dump
echo "OK: /backups/*_$ts.dump"

# backup-mywms.sh
#!/usr/bin/env bash
set -euo pipefail
ts=$(date +%F_%H%M%S)
pg_dump -U postgres -d mywms -Fc -f /backups/mywms_$ts.dump

# backup-reportserver.sh
#!/usr/bin/env bash
set -euo pipefail
ts=$(date +%F_%H%M%S)
pg_dump -U postgres -d reportserver -Fc -f /backups/reportserver_$ts.dump

# restore-mywms.sh
#!/usr/bin/env bash
set -euo pipefail
dump="$1"
pg_restore -U postgres -d mywms --clean --if-exists "$dump"
```

---

13) Alternative Dockerfile.wf22

```
FROM eclipse-temurin:11-jre
ADD wildfly-22.0.1.Final /opt/wildfly
ENV JBOSS_HOME=/opt/wildfly
WORKDIR ${JBOSS_HOME}
COPY standalone.xml ${JBOSS_HOME}/standalone/configuration/standalone.xml
EXPOSE 8080 9990
ENTRYPOINT ["bash","-lc","${JBOSS_HOME}/bin/standalone.sh -b 0.0.0.0 -bmanagement 0.0.0.0"]
```

---

14) Examples & templates

```
# pgAdmin access
# URL: http://<host>:${PGADMIN_HTTP}
# Host: pg, Port: 5432, User: postgres, Pass: from .env
```

---

15) Repository, Git & CI/CD

```
git init
git add .
git commit -m "Initial mywms-stack setup"

name: build-reportserver
on: [push]
jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: docker/login-action@v3
        with:
          registry: ghcr.io
          username: ${{ github.actor }}
          password: ${{ secrets.GITHUB_TOKEN }}
      - uses: docker/build-push-action@v5
```

```
  with:
    context: ./reportserver
    push: true
    tags: ghcr.io/ORG/reportserver:latest
```

---

16) README.md template

```
# myWMS Stack (PostgreSQL 16.9 + WildFly 22 + ReportServer CE)

> Production-ready dockerized stack for myWMS and ReportServer.
> This repository contains compose files, Dockerfiles, configs and helper scripts.

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## Architecture
- **PostgreSQL 16.9** — databases: `mywms`, `reportserver`.
- **myWMS on WildFly 22 (JDK 11)** — local WF tree, `standalone.xml`, and `deployments/` with
`mywms.ear` + `postgresql-42.2.19.jar`.
- **ReportServer CE (Tomcat 9 + JDK 11)** — unpacked `webapp/reportserver`, externalized configs in
`reportserver/config/`.
- **Startup order:** `postgres` → (optional `rs_init`) → `mywms` + `reportserver`.

## Prerequisites
- Docker ≥ 24, Docker Compose v2
- Ubuntu 20.04/22.04/24.04 (or compatible)
- RAM: 8–16 GB, CPU: 2–4 vCPU recommended
```

- Host paths created: `/srv/pgadata`, `/srv/pgbackup` (owned by the Docker user)

Quick start
```bash
cp .env.example .env
mkdir -p /srv/pgadata /srv/pgbackup && sudo chown -R $USER:$USER /srv/pgadata /srv/pgbackup

# 1) PostgreSQL
docker compose up -d postgres

# 2) (Optional) Import ReportServer schema once
docker compose run --rm rs_init

# 3) App containers
docker compose up -d mywms reportserver

# Health checks / logs
docker compose ps
docker compose logs -f postgres
docker compose logs -f mywms
docker compose logs -f reportserver
```

Environment variables
Key	Default	Description
`POSTGRES_USER`	`postgres`	Superuser
`POSTGRES_PASSWORD`	`postgres`	Superuser password
`MYWMS_DB`	`mywms`	myWMS database name
`MYWMS_USER`	`mywms`	myWMS DB user
`MYWMS_PASS`	`mywms`	myWMS DB password
`RS_DB`	`reportserver`	ReportServer DB name
`RS_USER`	`reportserver`	ReportServer DB user
`RS_PASS`	`reportserver`	ReportServer DB password
`PG_PORT`	`5432`	Host port for PostgreSQL
`MYWMS_HTTP`	`8080`	Host port → WF HTTP
`MYWMS_MGMT`	`9990`	Host port → WF management
`RS_HTTP`	`8085`	Host port → Tomcat
`PGADMIN_HTTP`	`5050`	(If using pgAdmin container)

````
| `PG_HOST_DATA` | `/srv/pgadata` | Host volume for PGDATA |
| `PG_HOST_BACKUP` | `/srv/pgbackup` | Host volume for dumps |

## Directory layout
```
mywms-stack/
├─ .env
├─ docker-compose.yml
├─ postgres/
│  ├─ conf/{pg_hba.conf, postgresql.conf}
│  └─ init/{01-create-db-users.sql, 02-load-mywms.sh, rs_post_init.sql}
├─ mywms/
│  ├─ Dockerfile
│  ├─ standalone.xml
│  └─ deployments/{mywms.ear, postgresql-42.2.19.jar}
└─ reportserver/
   ├─ Dockerfile
   ├─ webapp/reportserver/...
   └─ config/{persistence.properties, reportserver.properties}
```

## Startup & shutdown
```bash
# Up
docker compose up -d postgres
docker compose run --rm rs_init    # once
docker compose up -d mywms reportserver

# Down (reverse order)
docker compose stop reportserver mywms
docker compose stop postgres
```

## PostgreSQL schema import for ReportServer
The `rs_init` one-shot service runs the official DDL:
- `reportserver/webapp/reportserver/ddl/reportserver-RS4.7.7-6117-schema-PostgreSQL_CREATE.sql`
- plus grants from `postgres/init/rs_post_init.sql`

Re-run on a fresh database if needed:
````

```bash
docker compose run --rm rs_init
```

Updating `mywms.ear`
```bash
cp NEW_mywms.ear mywms/deployments/mywms.ear
touch mywms/deployments/mywms.ear.dodeploy
docker compose restart mywms
```

Backups & restore
Use scripts under `postgres/backup/` (mounted to `/backups` in the PG container).
```bash
# Full example
docker exec -it pg bash -lc '/backups/backup-all.sh'
# Restore mywms
docker exec -it pg bash -lc '/backups/restore-mywms.sh /backups/mywms_YYYY-MM-DD_HHMMSS.dump'
```

Troubleshooting
- Verify `.env` and `docker-compose.yml` values.
- If RS says “permission denied” on tables, re-apply grants (`postgres/init/rs_post_init.sql`).
- For Jasper Cyrillic PDFs: make sure `fonts/ttf/DejaVuSans*.ttf`, `fonts/dejavu.xml`, and Jasper extension properties exist inside the container; then restart RS and clear Tomcat work/temp.

License & disclaimer
This example is provided **AS IS**, without warranties. Review and harden before production (users/roles, `pg_hba.conf`, secrets, network policies, backups, monitoring).